The role of protostellar jet orientation in feedback on molecular clouds

Alice C. Quillen[1,*] and William Gonzalez[†]

[1] *Department of Physics and Astronomy, University of Rochester, Rochester, NY, 14627, USA*

(Dated: October 28, 2026)

Looking at starforge: Examination of starforge simulations show that sink particle angular momentum direction changes in a disk tend to happen during epochs of low accretion rate. However, sink particle angular momentum does not tend to remain fixed. The mass weighted output momentum of outflows over the lifetime of the massive is probably complex. By looking at the number of nearby bound sink particles we compare the role of variations in accretion direction to that from binaries. A rough model suggests that the length scale for energy injection due to jet feedback is sensitive to jet orientation variations.

CONTENTS

I. Introduction 1
A. Stuff 2

References 2

I. INTRODUCTION

While accreting, stars of all masses emit bipolar outflows which can be in the form of jets or winds [13]. These outflows impact the ambient molecular cloud causing turbulence, open cavities [10] and affect the star formation efficiency of the cloud [4, 9]. The positions of bright knots from emitting shocks in a protostellar jet may not line on a straight line, rather they can lie on a curved, S-shaped or corrugated locus [11, 12]. A large variation in jet orientation angle can be due to to precession [1, 8] if the jet's stellar source is part of a binary system. Precession is caused by the torque exerted on the source's disk [15]. If a jet precesses then it is less likely to puncture its host molecular cloud, and could even be decelerated [5]. Rubinstein et al. [14] interpreted proper motions from a protostar's jet in terms of a jet that is both decelerating and precessing. In that case feedback onto the host molecular cloud would be less efficient. The fraction of momentum that is injected into the host molecular cloud could be sensitive to jet direction and consequently on whether or not young stars are in binary systems while they are accreting.

Another way that jet direction can vary is if the angular rmomentum of accreted material from the host molecular cloud varies as a function of time.

Recent simulations of star formation have incorporated more physical processes including MHD, radiation, etc as well as feedback.

Do we think angular momentum variations due to accretion variations are more important than those due to nearby stars?

We should show the distribution of sink particle ejected momentum directions weighted by mass. This would be computed via recipe from simulation output because it is probably not recorded.

To get the torque, we would require a the moment of inertia matrix of the sink (disky) and the quadrupole moment of the gravitational field. We would have to make an assumption about the moment of inertia matrix of the sink. We have a size and an angular momentum.

We could evaluate the quadrupolar moment of the gravitational field by using all the mass in the simulation.

The precession rate of a jet about a primary of mass M_p due to a secondary of mass M_s

$$\omega_{\rm precession} = -\frac{15}{32} q \left(\frac{R}{D}\right)^3 \cos\delta \Omega_R \qquad (1)$$

with $q = M_s/M_p$ the ratio of secondary to primary mass. Here $\Omega_R \equiv \sqrt{GM_p/R^3}$ with R disk radius. Here D is approximately the semi-major axis of the binary. The angle δ is the inclination of disk plane w.r.t to the binary orbit. It is also the angle between the disk angular momentum and the binary orbit angular momentum. This formula comes from Terquem et al. [15].

Starforge [6] takes into account the angular momentum of accreted gas and sink particles are described with an angular momentum vector that is a function of time and determines the direction of emitted jets. However starforge currently neglects tidal torques and so does not exhibit jet precession due to the precession of a binary. Though stars formed in starforge do exhibit multiplicity [7].

The total momentum output per unit time for a protostar is

$$\dot{P}_J = v_{\rm jet}\dot{M}_{\rm jet} = f_K f_w \dot{M}_{\rm acc}\sqrt{GM_*/R_*} \qquad (2)$$

Starforge adopts $f_w = 0.3$ and $f_K = 0.3$, similar to the values used by Cunningham et al. [3]. The paramater f_w

* aquillen@ur.rochester.edu
† wgonzal3@u.rochester.edu

is the fraction of mass ejected compared to accreted and f_K is the jet velocity in Keplerian units.

About half of the stars formed in the STARFORGE calculations are members of multiple systems, with overall statistics that are in reasonable agreement with the ob- servedmultiplicityandcompanionstarfractionswhentheperiastron is larger than the softening radius [7]. Mass ratio peaks at about 0.2 and semi-major axis peaks at 20 AU where the softening is a problem. Hence there are lots of close binaries.

Current simulation have a top heavy IMF and massive stars tend to increase in mass too fast apparently.

A. Stuff

To support the assumption that we need only take into account the time that the jet is on, we note that simulations show that jets heads stop advancing after the jet source is extinguished, though the cavity opened in the molecular cloud can continue to slowly expand afterwards [2] and as proposed from observations by Quillen et al. [10]. Short lived jets are subsumed by turbulent motions [2].

Suppose a jet has density ρ and velocity v_j and a solid angle $\Omega(r)$. Then

$$\dot{m} = \rho(r) v_j \Omega(r) r^2$$

is the mass outflow rate. That implies that

$$\rho(r) = \frac{\dot{m}}{v_j \Omega(r) r^2}.$$

Assume that v_j is constant. To estimate a shock advance speed take

$$p = \rho(r) v_j v_a,$$

where p is the pressure of the ISM. This gives

$$v_a(r) = \frac{p\Omega(r) r^2}{\dot{m}}$$

and is independent of v_j and increases with radius which does not make sense unless Ω cancels out the r^2. We assume that $A = \Omega r^2$ is fixed. The jet does not expand and has a constant thickness.

We could take a jet advance speed

$$v_a \sim \frac{pA}{\dot{m}} \tag{3}$$

units ok. This gives an advance speed velocity that only depends on the mass outflow rate $\dot{m}$ (ignoring variations in A, p. The work done on the ISM per unit time is $v_a pA$.

Suppose that the jet is on a length of time T. The total energy dumped into the ISM during the burst is

$$U \sim v_a pAT \sim \frac{p^2 A^2 T}{\dot{m}}. \tag{4}$$

The distance the jet advances is $D = v_a T$. The work done per unit distance is constant and the energy dumped per unit distance is then $u \sim \frac{v_a pTA}{v_a T} \sim pA$ and is approximately independent of jet physics. However, the length over which is this energy is dumped is sensitive to jet velocity and time. The mean length where energy is dumped is $v_a T/2$.

If the jet experiences a series of bursts each with $\dot{m}_i$ and time T_i. Then the total energy dumped is

$$U_{\rm tot} \sim \sum_i \frac{p^2 A^2 T_i}{\dot{m}_i}.$$

If we weighted the distances by the energy then the average distance of energy dumping would be

$$\bar{L} \sim \sum_i \frac{v_{a,i} T_i}{2} \times \frac{p^2 A^2 T_i}{\dot{m}_i} \frac{1}{U_{\rm tot}}$$

$$\bar{L} \sim \sum_i \frac{p^3 A^3 T_i^2}{2\dot{m}_i^2} \frac{1}{U_{\rm tot}}. \tag{5}$$

$$\bar{L} \sim \frac{pA}{2} \frac{\sum_i T_i^2/\dot{m}_i^2}{\sum_i T_i/\dot{m}_i} \tag{6}$$

The longer the time of each burst, the longer is $\bar{L}$. This gives us at least an idea of the effect we want. But this model is inconsistent with S-shaped jets. Maybe not completely as some fraction of the energy continues to where the jet was rather than is pointing. The fact that work is done nearer by when the jet is S shaped is probably correct.

We could add a fudge factor for the effective length of time? If the jet is precessing we say that T_i is the length of time it takes the jet to change orientation angle by A which in this model is dependent on r.

We could allow A to depend on r or we just choose a distance to let us set T_i for each accretion event.

[1] Anglada, G., López, R., Estalella, R., Masegosa, J., Riera, A., Raga, A.C., 2007. Proper motions of the jets in the region of hh 30 and hl/xz tau: Evidence for a binary

exciting source of the hh 30 jet. The Astronomical Journal 133, 2799. URL: https://doi.org/10.1086/517493, doi:doi:10.1086/517493.

[2] Cunningham, A.J., Frank, A., Carroll, J., Blackman, E.G., Quillen, A.C., 2009. Protostellar Outflow Evolution in Turbulent Environments. Astrophysical Journal 692, 816–826. doi:doi:10.1088/0004-637X/692/1/816, arXiv:0804.4197.

[3] Cunningham, A.J., Klein, R.I., Krumholz, M.R., McKee, C.F., 2011. Radiation-hydrodynamic Simulations of Massive Star Formation with Protostellar Outflows. Astrophysical Journal 740, 107. doi:doi:10.1088/0004-637X/740/2/107, arXiv:1104.1218.

[4] Cunningham, A.J., Krumholz, M.R., McKee, C.F., Klein, R.I., 2018. The effects of magnetic fields and protostellar feedback on low-mass cluster formation. Monthly Notices of the Royal Astronomical Society 476, 771–792. doi:doi:10.1093/mnras/sty154, arXiv:1709.01277.

[5] Fendt, C., Yardimci, M., 2022. Curved jet motion. i. orbiting and precessing jets. The Astrophysical Journal 933, 71. URL: https://doi.org/10.3847/1538-4357/ac7145, doi:doi:10.3847/1538-4357/ac7145.

[6] Grudić, M.Y., Guszejnov, D., Hopkins, P.F., Offner, S.S.R., Faucher-Giguère, C.A., 2021. STARFORGE: Towards a comprehensive numerical model of star cluster formation and feedback. Monthly Notices of the Royal Astronomical Society 506, 2199–2231. doi:doi:10.1093/mnras/stab1347, arXiv:2010.11254.

[7] Guszejnov, D., Raju, A.N., Offner, S.S.R., Grudić, M.Y., Faucher-Giguère, C.A., Hopkins, P.F., Rosen, A.L., 2023. Effects of the environment on the multiplicity properties of stars in the STARFORGE simulations. Monthly Notices of the Royal Astronomical Society 518, 4693–4712. doi:doi:10.1093/mnras/stac3268, arXiv:2208.02844.

[8] Masciadri, E., Raga, A.C., 2002. Herbig-Haro Jets from Orbiting Sources. Astrophysical Journal 568, 733–742. doi:doi:10.1086/338767.

[9] Matzner, C.D., McKee, C.F., 2000. Efficiencies of low-mass star and star cluster formation. The Astrophysical Journal 545, 364. URL: https://doi.org/10.1086/317785, doi:doi:10.1086/317785.

[10] Quillen, A.C., Thorndike, S.L., Cunningham, A., Frank, A., Gutermuth, R.A., Blackman, E.G., Pipher, J.L., Ridge, N., 2005. Turbulence Driven by Outflow-blown Cavities in the Molecular Cloud of NGC 1333. Astrophys. J. 632, 941–955. doi:doi:10.1086/444410, arXiv:astro-ph/0503167.

[11] Raga, A.C., Esquivel, A., Velázquez, P.F., Cantó, J., Haro-Corzo, S., Riera, A., Rodríguez-González, A., 2009. Mirror and Point Symmetries in a Ballistic Jet from a Binary System. Astrophysical Journal Letters 707, L6–L11. doi:doi:10.1088/0004-637X/707/1/L6.

[12] Raga, A.C., Velázquez, P.F., Cantó, J., Masciadri, E., 2002. The time-dependent ejection velocity histories of HH 34 and HH 111. Astronomy and Astrophysics 395, 647–656. doi:doi:10.1051/0004-6361:20021180.

[13] Richer, J.S., Shepherd, D.S., Cabrit, S., Bachiller, R., Churchwell, E., 2000. Molecular Outflows from Young Stellar Objects, in: Mannings, V., Boss, A.P., Russell, S.S. (Eds.), Protostars and Planets IV, p. 867. doi:doi:10.48550/arXiv.astro-ph/9904097, arXiv:astro-ph/9904097.

[14] Rubinstein, A.E., Karnath, N., Quillen, A.C., Federman, S., Green, J.D., Chambers, E.T., Watson, D.M., Megeath, S.T., 2023. HOPS 361-C's Jet Decelerating and Precessing through NGC 2071 IR. Astrophys. J. 948, 39. doi:doi:10.3847/1538-4357/acc401, arXiv:2211.12599.

[15] Terquem, C., Eislöffel, J., Papaloizou, J.C.B., Nelson, R.P., 1999. Precession of Collimated Outflows from Young Stellar Objects. Astrophysical Journal Letters 512, L131–L134. doi:doi:10.1086/311880, arXiv:astro-ph/9812155.